

Answer **all** questions in the spaces provided.

1. (a) State, in terms of electrons, the difference between ionic and covalent bonding. [2]

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- (b) When magnesium reacts with oxygen, the ionic compound magnesium oxide is formed.

- (i) The outer electrons of magnesium and oxygen are shown below. Draw arrows to show the transfer of electrons when magnesium oxide is formed. [1]



- (ii) State the charges on the ions formed [2]

Mg O



- (b) (i) State **two** advantages of the materials used in 2015 over those used previously. [2]

1.

2.

- (ii) State **one** disadvantage of the materials used in 2015 over those used previously. [1]

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- (c) GRP is a common composite material, in which glass fibres are mixed with epoxy resin.

- (i) A GRP panel contains a volume of 300 cm^3 of epoxy resin.

Use the equation: [3]

$$\text{mass} = \text{density} \times \text{volume}$$

to calculate the mass of the resin.

mass = g

- (ii) I. State the strength of glass fibre in N/cm^2 . [2]

$$(1 \text{ MPa} = 100 \text{ N/cm}^2)$$

strength = N/cm^2

- II. The cross-sectional area (csa) of a glass fibre is 0.00015 cm^2 .

Use your answer in part (c)(ii)I. and the equation:

$$\text{force} = \text{strength} \times \text{csa}$$

to calculate the force required to break it. [2]

force = N

